Road Accident and Traffic Safety Analysis

Introduction / Project Overview

Road traffic accidents are a major global public safety concern, causing injuries, fatalities, and economic losses every year. Understanding the patterns and causes of accidents can help policymakers, transport authorities, and drivers improve road safety.

This project aims to analyze road accident data using Python to identify trends, patterns, and factors that may influence accident occurrence and severity. Through data cleaning, exploratory data analysis, and visualization techniques, the project will generate insights that can help better understand traffic safety risks.

Dataset Source

The dataset used in this project is the US Accidents dataset, obtained from Kaggle. This dataset contains detailed information about traffic accidents across multiple states in the United States, including accident severity, time of occurrence, weather conditions, and location data.

Due to the large size of the original dataset, a sample of the dataset was used for analysis.

Number of Rows

The sampled dataset contains over 48,214 and multiple (46) variables describing accident conditions and characteristics

Dataset Description

The dataset records traffic accidents and includes several variables that allow analysis of accident patterns, severity, and contributing environmental factors.

Some key variables in the dataset include:

Variable	Description
Severity	Indicates the severity level of the accident
Start time	Date and time when the accident occurred
City	City where the accident occurred
State	State where the accident occurred
Weather condition	Weather conditions during the accident
Temperature	Temperature at the time of the accident
Visibility	Visibility distance during the accident
Humidity	Humidity level
Wind speed	Wind speed during the accident
description	Description of the accident event

These variables provide useful information for analyzing how environmental and temporal factors affect road accidents.

Research Questions

The following research questions will guide the analysis in this project:

1. Which locations or states experience the highest number of road accidents?
2. What time of day has the highest accident frequency?
3. How do weather conditions influence accident severity?

4. Which months or seasons have the highest number of accidents?
5. What is the distribution of accident severity levels in the dataset?

Expected Outcome

The project is expected to identify meaningful patterns in road accident data, including high-risk locations, peak accident periods, and environmental conditions associated with accidents. These insights may help improve awareness of traffic safety issues and highlight potential areas for safety improvement.

3. Data Cleaning and Preparation

The dataset used in this study required several preprocessing steps to ensure it was suitable for analysis. Raw datasets often contain inconsistencies such as missing values, duplicate records, and incorrect data formats, which can negatively affect the accuracy of results. Therefore, data cleaning was an essential step before performing any analysis.

First, the dataset was inspected to understand its structure, including the number of rows, columns, and data types. This helped identify potential issues such as missing values and incorrect formats. Duplicate records were then removed to prevent repeated data from influencing the analysis results.

A key step in the cleaning process was handling the Start_Time variable. Initially, this column was stored as a string, which limited its usability for time-based analysis. It was converted into a datetime format to enable extraction of meaningful features such as year, month, hour, and day of the week. These new variables were crucial for analyzing temporal patterns in accidents.

Missing values were handled carefully. Instead of removing all rows with missing values, only critical columns such as time and severity were prioritized. This ensured that the dataset retained sufficient data for meaningful analysis while maintaining accuracy.

Additionally, new variables were created from the datetime column. These included:

- Year: to analyze trends over time
- Month: to study seasonal patterns
- Hour: to examine time-of-day effects
- Day: to understand weekly trends

These transformations improved the dataset by making it more structured and analysis-ready. After cleaning, the dataset was consistent, reliable, and suitable for both exploratory and advanced analysis.

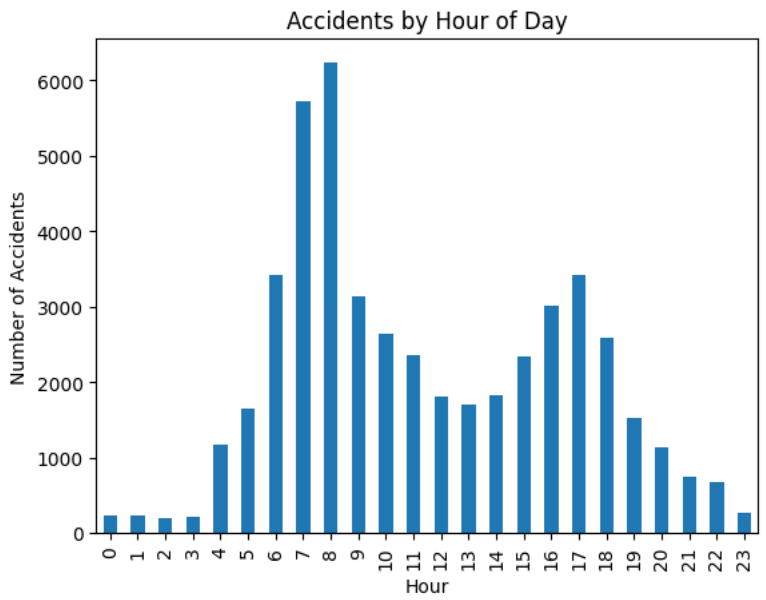
4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns, trends, and distributions within the dataset. This stage focused on visualizing the data and identifying key characteristics of road accidents.

4.1 Accident Counts by Hour of the Day

The analysis of accident counts by hour revealed that accidents are not evenly distributed throughout the day. Instead, there are clear peaks during specific hours, particularly in the morning and evening.

These peak periods correspond to typical commuting hours when traffic volume is significantly higher. During these times, roads are more congested, increasing the likelihood of collisions. The findings highlight the strong relationship between traffic density and accident occurrence.



4.2 Top States with Most Accidents

Although a balanced dataset with 1,000 samples per state was created to ensure equal representation, it was found to be less suitable for certain types of analysis. This is because limiting each state to the same number of observations removes the natural differences in accident frequency across states.

In reality, some states experience significantly more accidents than others due to factors such as population size, traffic volume, and urbanization. By forcing each state to have the same number of records, the dataset becomes artificially balanced, which can distort the true patterns in the data.

To address this limitation, the original cleaned dataset containing approximately 8.7 million records was used for analyses that required an accurate representation of accident distribution across states. This allowed for a more realistic understanding of how accidents vary geographically.

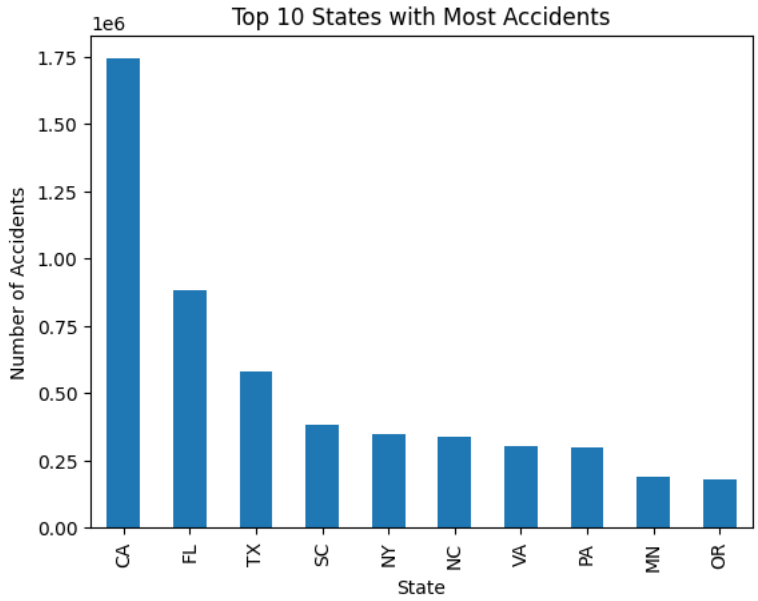
Therefore, two datasets were used in this study:

- A balanced sampled dataset for general analysis and visualization
- The full dataset for analyses involving actual accident distribution by state

This approach ensured both computational efficiency and analytical accuracy.

The analysis of accident distribution by state showed that a small number of states account for a large proportion of total accidents. This indicates that accident occurrences are geographically concentrated rather than evenly spread across all regions.

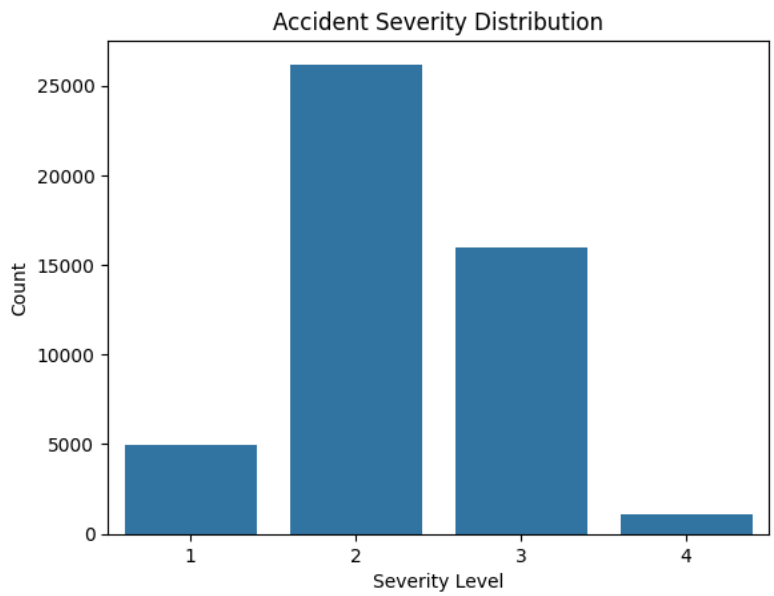
Several factors may explain this pattern, including population size, urbanization, traffic density, and road infrastructure. States with larger cities and higher vehicle usage are more likely to experience higher accident rates. This analysis helps identify high-risk regions that may require targeted safety interventions.



4.3 Accident Severity Distribution

The distribution of accident severity levels shows that most accidents fall within moderate categories, while fewer cases are classified as highly severe. This suggests that although accidents occur frequently, extreme outcomes are relatively less common.

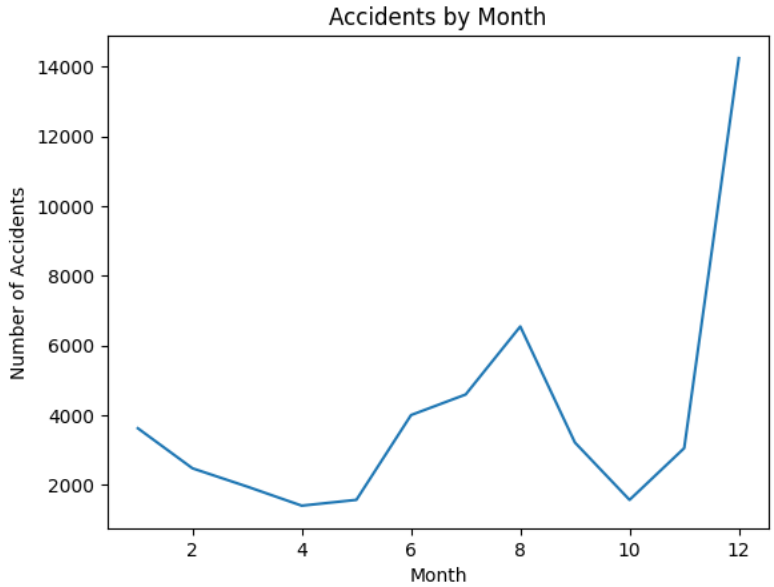
This pattern may indicate that many accidents happen at lower speeds or under conditions where damage is limited. However, even moderate accidents can have significant impacts, emphasizing the need for continuous safety improvements.



4.4 Accidents by Month

The monthly analysis of accidents revealed noticeable variations throughout the year. Certain months recorded higher accident frequencies compared to others, suggesting the influence of seasonal factors.

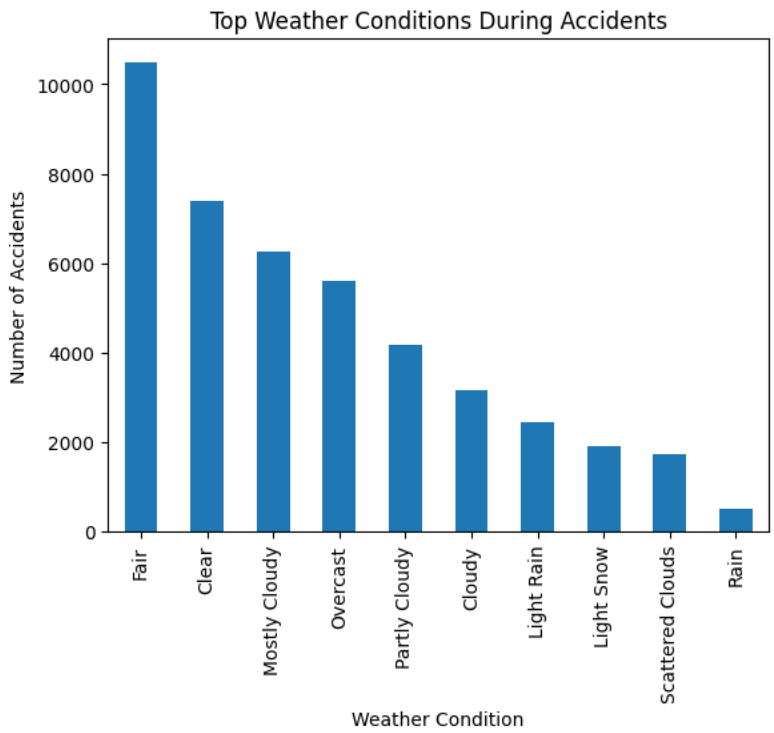
These variations may be linked to weather conditions, holiday travel, and changes in driving behavior. For example, increased travel during holiday periods or adverse weather conditions during certain seasons may contribute to higher accident rates.



4.5 Weather Conditions During Accidents

The analysis of weather conditions showed that a large number of accidents occur during normal weather conditions such as clear or cloudy skies. This is expected, as such conditions occur more frequently.

However, adverse weather conditions such as rain, fog, and reduced visibility were also associated with accidents. These conditions can impair driver visibility and reduce road traction, increasing the risk of accidents. This highlights the importance of cautious driving during unfavorable weather conditions.



5. Advanced Analysis

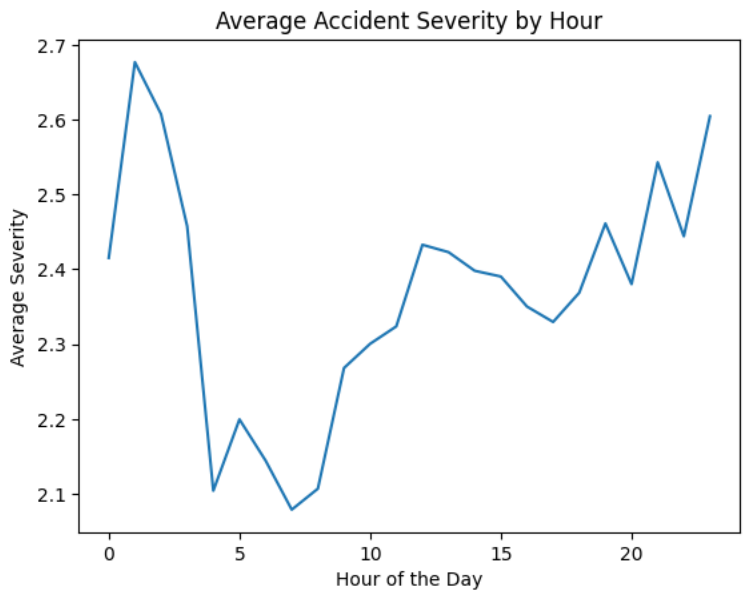
In this section, a deeper analysis was conducted to examine the relationships between different variables in the dataset. Unlike exploratory analysis, which focuses on describing patterns, advanced analysis aims to understand how different factors influence accident severity and occurrence.

5.1 Accident Severity by Hour of the Day

The analysis of accident severity across different hours of the day revealed that severity levels are not constant throughout the day. Certain hours tend to have slightly higher average severity compared to others.

This variation may be influenced by several factors. For example, late-night and early-morning hours often have lower visibility and reduced driver alertness, which can lead to more severe accidents. In contrast, while rush hours have a higher number of accidents, they may involve slower-moving traffic, potentially resulting in less severe collisions.

This finding highlights that the timing of accidents not only affects frequency but also severity, making it an important factor in road safety planning.

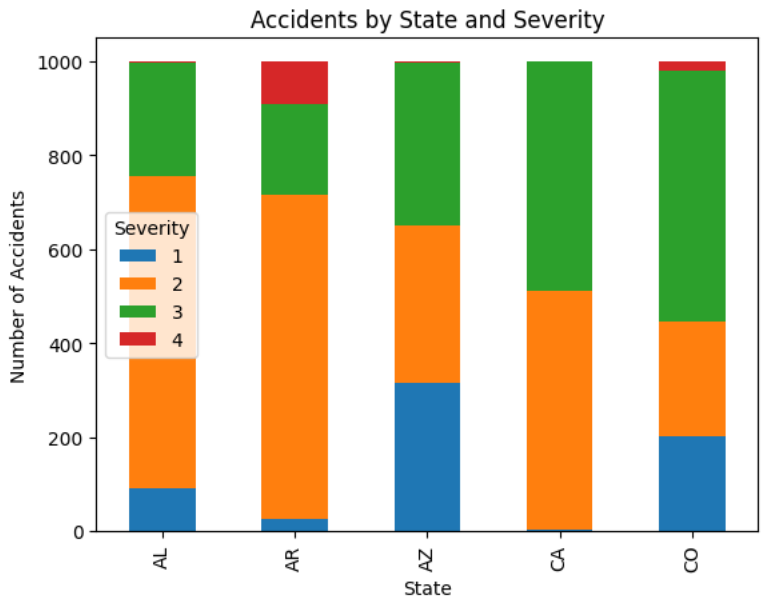


5.2 Accidents by State and Severity

The relationship between accident location and severity was also examined by comparing severity levels across different states. The results show that not only do accident frequencies vary by state, but the severity of accidents also differs across regions.

Some states exhibit higher average severity levels, which may be attributed to factors such as road design, traffic speed, enforcement of traffic laws, and emergency response systems. Rural areas, for example, may experience more severe accidents due to higher driving speeds and longer emergency response times.

This analysis suggests that geographical location plays a significant role in determining both the likelihood and severity of accidents, emphasizing the need for region-specific safety measures.

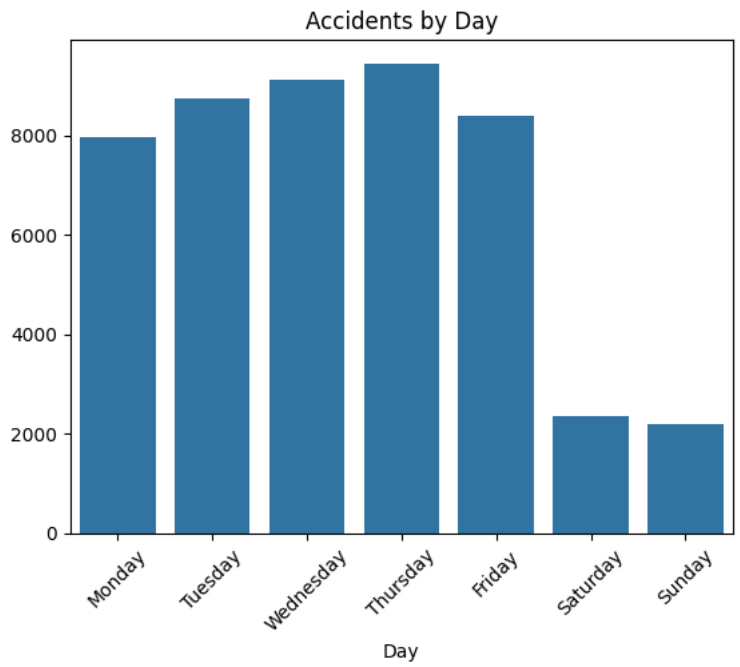


5.3 Accident Severity by Day of the Week

The analysis of accident severity across days of the week revealed noticeable variations linked to human activity patterns. Weekdays generally experience a higher volume of traffic due to work-related commuting, while weekends may involve different driving behaviors such as leisure travel.

Although the number of accidents may vary, the severity of accidents can also differ depending on the day. For instance, weekends may involve higher-speed travel or nighttime driving, which can increase the risk of severe accidents.

This indicates that behavioral patterns and daily routines influence not only how often accidents occur but also how severe they are.

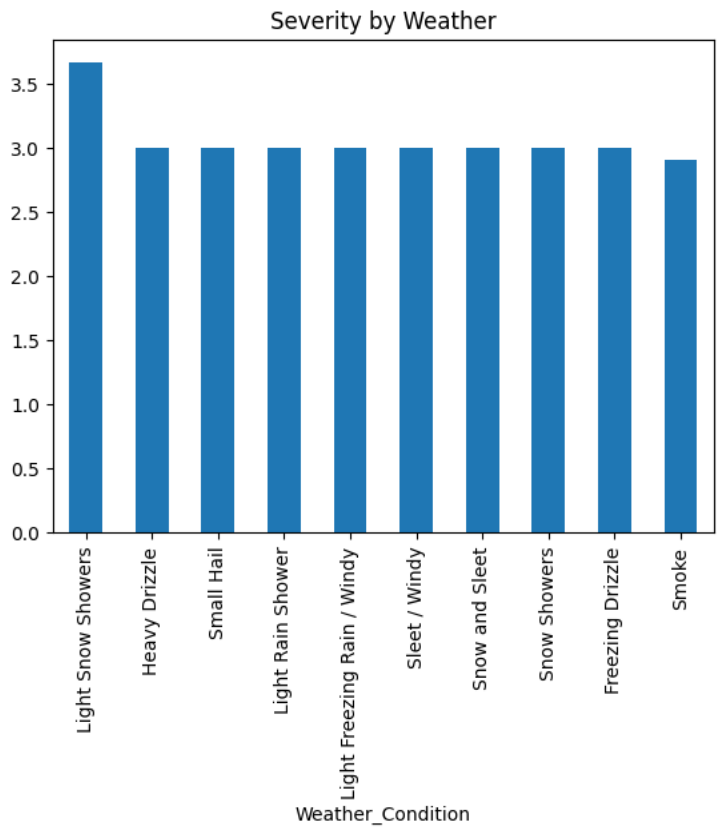


5.4 Accident Severity by Weather Conditions

Weather conditions were found to have a significant impact on accident severity. While many accidents occur under normal weather conditions due to their frequency, adverse conditions such as rain, fog, and low visibility are associated with higher severity levels.

Poor weather affects driving conditions by reducing visibility, decreasing tire traction, and increasing stopping distances. These factors make it more difficult for drivers to react quickly and safely, thereby increasing the likelihood of more serious accidents.

This analysis demonstrates that environmental factors play a critical role in accident outcomes, particularly in determining severity.



5.5 Combined Influence of Multiple Factors

The analysis also suggests that accident severity is rarely influenced by a single factor. Instead, it is often the result of a combination of elements such as time, weather, location, and human behavior.

For example, a severe accident is more likely to occur during nighttime under poor weather conditions in high-speed areas. This highlights the importance of considering multiple variables when analyzing road safety data.

Overall, the advanced analysis provides a more comprehensive understanding of the factors that contribute to accident severity, moving beyond simple counts to reveal deeper patterns and relationships.

6. Key Insights

Based on the analysis conducted in this study, several important insights were identified regarding the patterns and causes of road accidents.

- **Concentration of Accidents in Specific Locations**

The analysis revealed that accidents are not evenly distributed across all states. Instead, a small number of states account for a large proportion of total accidents. This suggests that certain regions experience higher traffic density, urbanization, or risk factors that contribute to increased accident occurrence with California being the most affected

- **Peak Hours Significantly Influence Accident Occurrence**

Accidents were found to be more frequent during specific hours of the day, particularly during morning and evening peak periods. These times are associated with heavy traffic congestion, which increases the likelihood of collisions.

- **Severity Varies with Time of Day**

The severity of accidents is influenced by the time at which they occur. Certain hours, especially late at night or early in the morning, tend to have higher severity levels. This may be due to reduced visibility, driver fatigue, or higher driving speeds.

- **Weather Conditions Increase Risk of Severe Accidents**

Adverse weather conditions such as rain, fog, and low visibility were found to contribute to more severe accidents. These conditions reduce driving safety and increase the difficulty of controlling vehicles, leading to more serious outcomes.

- **Human Behavior and Daily Patterns Play a Key Role**

The variation in accident patterns across days of the week highlights the role of human behavior. Weekday commuting and weekend travel patterns influence both the frequency and severity of accidents.

- **Accidents Are Influenced by Multiple Interacting Factors**

One of the most important insights from this study is that accidents are rarely caused by a single factor. Instead, they result from the interaction of multiple variables, including time, location, weather, and driver behavior.

This emphasizes the need for a comprehensive approach to road safety that considers all contributing factors.

7. Recommendations

Based on the findings from the analysis, several practical recommendations can be proposed to reduce the frequency and severity of road accidents. These recommendations focus on improving road safety through better traffic management, infrastructure, and public awareness.

- **Improve Traffic Management During Peak Hours**

Since accidents are more frequent during morning and evening rush hours, it is important to implement strategies that reduce congestion during these periods. This can include the use of intelligent traffic systems, improved traffic signal coordination, and better road planning.

Reducing congestion not only lowers the likelihood of accidents but also improves overall traffic flow and efficiency.

- **Target High-Risk Locations**

The analysis showed that certain states and regions experience a higher number of accidents. Authorities should prioritize these high-risk areas by improving road infrastructure, increasing signage, and enhancing road lighting.

Additionally, regular monitoring and maintenance of roads in these areas can help prevent accidents caused by poor road conditions.

- **Enhance Road Safety in Adverse Weather Conditions**

Since weather conditions significantly affect accident severity, measures should be taken to improve safety during unfavorable conditions. This can include installing better drainage systems, improving road surface quality, and providing real-time weather alerts to drivers.

Drivers should also be encouraged to adjust their speed and driving behavior during adverse weather conditions.

- **Increase Public Awareness and Driver Education**

Human behavior plays a major role in road accidents. Therefore, public awareness campaigns should be implemented to educate drivers about safe driving practices, especially during high-risk periods such as peak hours and poor weather conditions.

Driver education programs can emphasize the importance of maintaining safe speeds, avoiding distractions, and following traffic rules.

- **Implement Data-Driven Decision Making**

The use of data analytics in this study demonstrates the value of data-driven decision-making. Authorities should continue to collect and analyze accident data to identify trends and implement targeted interventions.

Using data to guide policy decisions ensures that resources are allocated efficiently and that safety measures are based on evidence.

- **Improve Emergency Response Systems**

In cases where accidents occur, the speed and effectiveness of emergency response can significantly impact outcomes. Improving emergency services, reducing response times, and ensuring proper coordination between agencies can help reduce fatalities and severity.

8. Conclusion

This study analyzed road accident data to identify patterns, trends, and factors influencing accident occurrence and severity. The findings revealed that accidents are influenced by a combination of time, location, weather conditions, and human behavior.

The analysis showed that accidents are more frequent during peak traffic hours and are concentrated in specific regions. Additionally, factors such as adverse weather conditions and certain times of the day contribute to increased accident severity.

These findings highlight the importance of adopting a comprehensive approach to road safety that addresses multiple contributing factors. By implementing targeted interventions, improving infrastructure, and promoting safe driving practices, it is possible to reduce both the frequency and severity of accidents.

Overall, this study demonstrates the value of data analysis in understanding complex real-world problems. The insights gained can support decision-making and contribute to the development of effective road safety strategies.

Future work could expand on this analysis by incorporating predictive modeling techniques and real-time data to further enhance accident prevention efforts.